

BrokerVerse

Out-of-the-box insurance broking platform

Solution Architecture · Document 03 of 11

Database Inventory

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Review and approval

Role	Name	Signature	Date
Solution architect			
DevOps / infrastructure lead			
Information security			
Business owner			

Conventions

- Statements of fact are derived from the code, configuration and database of the baseline above (file paths are given relative to the repository). Figures measured on the local test system are labelled **test data** or **indicative**.
- Targets, sizes, retention periods and thresholds that are not facts of the code are labelled **Recommended / to be confirmed by the business and DevOps**.
- Items marked **Gap** are defects or missing capabilities found while preparing this document.

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1 Introduction

1.1 Purpose and scope

This document is the inventory of the BrokerVerse database: every table with its domain, purpose, key columns, row count and size, number of indexes, retention class and owning backend module, followed by the indexes, sequences, extensions, functions, triggers and the application settings by group. It supports capacity planning (document 06), backup and archival (documents 09 and 10) and data-protection reviews. The design rationale is in document 02.

1.2 Source and currency of the figures

The inventory is generated from the catalogue of the local PostgreSQL 16 database `brokerverse` by `docs/architecture/tools/collect_db_inventory.py` (read-only catalogue queries and one `count(*)` per table) and the hand-maintained descriptions in `tools/table_catalog.py`.

Note: Row counts and sizes are those of the local **test database** after the end-to-end test run and the sample seed (`SEED_SAMPLE_DATA` on). They show relative weight, not production volumes. A production database starts with reference data only. Re-run the collector against UAT or production and rebuild the documents to refresh them (see `docs/architecture/README.md`).

Size is the total relation size in KB (table, TOAST and indexes; PostgreSQL allocates at least one 8 KB page per relation and index, so small tables show 16 to 64 KB). "Idx" is the number of indexes including the primary key.

1.3 Retention classes

Class	Meaning and default handling (periods in document 10)
Financial	Financial records: ledger, bills, receipts, payments, vouchers, commissions, remittances, debit notes, reinsurance settlements. Kept for the statutory record-keeping period; never deleted from the online system while the period runs; cancelled or reversed, not deleted.
Transaction	Business transaction records: leads, clients, quotations, policies, endorsements, claims, renewals, treaties, cessions, documents. Kept for the policy / claim life plus the regulatory period.
Reference	Reference data and configuration: masters, settings, product templates, roles, permissions, report catalogue, scheduled jobs. Kept while in use; changes audited.
Log / audit	Audit log, sign-in history, histories and field-level trails, job run history, notice logs. Kept for the audit period, then archived.
Temporary	Temporary and operational data: tokens, reset codes, e-mail outbox, notifications, work queues, generated report files. Purged by housekeeping after a short period.

2 Summary

2.1 Tables by domain

Domain	Tables	Rows (test data)	Size KB (test data)	Tables
Security and identity	9	1,047	712	login_history, password_history, password_resets, permissions, refresh_tokens, role_permissions, roles, user_roles, users
Platform services	13	2,327	1,608	agent_events, app_settings, audit_log, documents, email_outbox, job_queue, job_runs, master_records, master_types, notifications, scheduled_jobs, schema_migrations, sequences
Reference masters	16	232	704	banks, branches, cities, countries, coverages, currencies, districts, insurance_companies, policy_types, postal_codes, products, signatories, states, vehicle_brands, vehicle_models, vehicle_variants
Product configurator	4	73	280	product_components, product_risk_mappings, product_risk_sections, product_templates
Sales and policy	6	103	688	clients, endorsements, leads, policies, policy_payments, quotes
Claims	3	134	280	claim_field_changes, claim_history, claims
Renewals	7	57	416	renewal_activities, renewal_batch_policies, renewal_batches, renewal_notices, renewal_quotes, renewals, winback_campaigns
Billing, receipts and collections	6	156	432	collection_actions, collection_items, receipt_applications, receipt_lines, receipts, receivables
General ledger	5	531	520	accounting_periods, entry_matches, gl_accounts, journal_lines, journal_vouchers
Disbursements and petty cash	9	41	448	checkbooks, disbursements, invoice_lists, petty_cash_disbursements, petty_cash_funds, petty_cash_receipts, petty_cash_replenishments, petty_cash_request_lines, petty_cash_requests
Commission	2	63	152	commission_referrers, commissions
Remittance and direct bill	9	105	568	commission_debit_note_collections, commission_debit_note_lines, commission_debit_notes, direct_bill_items, remittance_approvals, remittance_delegations, remittance_items, remittance_lines, remittances
Reinsurance	7	37	320	cessions, reinsurance_bordereaux, reinsurance_exceptions, reinsurance_reconciliations, reinsurance_recoveries, reinsurance_treaties, reinsurers
Incentive	3	26	144	incentive_calculations, incentive_programs, incentive_results
Reporting	3	134	272	generated_reports, report_definitions, report_schedules
Total	102	5,066	7,544	

2.2 Tables with the most rows (test data)

Table	Rows (test data)	Total size KB	Average row bytes	Retention class
job_runs	961	248	145	Log / audit
audit_log	574	400	424	Log / audit
refresh_tokens	352	248	153	Temporary
login_history	351	168	169	Log / audit
master_records	317	384	385	Reference
journal_lines	286	152	180	Financial
app_settings	273	128	190	Reference
role_permissions	226	56	32	Reference

Table	Rows (test data)	Total size KB	Average row bytes	Retention class
gl_accounts	128	128	166	Reference
journal_vouchers	117	192	338	Financial
generated_reports	115	144	525	Temporary
claim_field_changes	63	80	132	Log / audit
claim_history	54	48	90	Log / audit
commissions	53	120	332	Financial
master_types	52	96	900	Reference

3 Table inventory

Key columns: PK primary key, UK unique constraint, FK foreign-key columns (first four), JSONB document columns. Rows and KB are test data.

Security and identity (9 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
login_history	Every sign-in attempt: user, IP, user agent, success, reason, method	PK id FK user_id	351	168	3	Log / audit	auth
password_history	Previous password hashes for the history rule (last N kept)	PK id FK user_id	15	48	2	Log / audit	auth
password_resets	One-time password reset codes (HMAC hash only), expiry and attempts	PK id FK user_id	0	16	1	Temporary	auth
permissions	Permission codes read:<module> / write:<module> (25 modules x 2)	PK id UK code	50	48	2	Reference	users
refresh_tokens	Refresh-token register (jti, family, expiry, rotation / revocation reason)	PK jti FK user_id	352	248	3	Temporary	auth
role_permissions	Grant of permissions to roles	PK role_id+permission_id	226	56	1	Reference	users
roles	Role catalogue (it-admin, ba, sales, underwriting, customer-services, claims, finance, agent, user-access-admin and custom roles)	PK id UK code	9	48	2	Reference	users
user_roles	Assignment of roles to users	PK user_id+role_id	22	32	1	Reference	users
users	User accounts: bcrypt password hash, status, lockout counter, 2FA (encrypted TOTP secret), token version	PK id UK username	22	48	2	Reference	users, auth

Platform services (13 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
agent_events	Agent calendar events / reminders	PK id FK user_id	0	16	1	Transaction	payments (open items)
app_settings	Configuration keys (JSONB value, group, label, type); every business parameter	PK key JSONB value	273	128	1	Reference	settings, system-settings
audit_log	Who did what to which record, with before / after JSON and IP	PK id JSONB before_data, after_data	574	400	2	Log / audit	all (lib/audit.js)
documents	Registry of stored files: storage key, name, type, size, owning entity, uploader	PK id UK storage_key	24	48	2	Transaction	uploads and file writers
email_outbox	Queued e-mails (to, subject, HTML body, status, attempts, error)	PK id	14	32	1	Temporary	lib/mailer.js
job_queue	Background work queue (renewal notice batches); status, progress, attempts	PK id JSONB payload, progress, result	0	24	2	Temporary	renewals
job_runs	Run history of scheduled and manual job runs (status, output, error)	PK id FK job_id JSONB output	961	248	1	Log / audit	jobs
master_records	Generic master records (JSONB data) for masters without their own table	PK id FK type_code JSONB data	317	384	4	Reference	masters
master_types	Master screen registry: fields, storage table, keys of each configurable master	PK code JSONB fields, unique_keys	52	96	1	Reference	masters
notifications	In-app notifications per user or per permission audience	PK id FK user_id	39	80	2	Temporary	notifications

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
scheduled_jobs	Cron jobs (code, cron, handler, params, enabled, last status)	PK id UK code JSONB params	9	48	2	Reference	schedules, jobs
schema_migrations	Applied migration files	PK name	32	32	1	Reference	db/migrate.js
sequences	Document-number counters per name and year (QT-, POL-, OR-, JV- ...)	PK name+period	32	72	1	Reference	all (next_number())

Reference masters (16 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
banks	Banks and mortgagees	PK id UK code JSONB attrs	6	48	2	Reference	masters
branches	Broker branches	PK id UK code JSONB attrs	3	48	2	Reference	masters
cities	Cities and municipalities	PK id FK state_id JSONB attrs	49	64	1	Reference	masters
countries	Countries	PK id UK code JSONB attrs	5	48	2	Reference	masters
coverages	BI / PD / PA cover options with amount and premium	PK id FK policy_type_id JSONB attrs	11	32	1	Reference	quotations
currencies	Currencies (base flag, decimals, exchange rate)	PK id UK code JSONB attrs	5	48	2	Reference	masters
districts	Districts / barangays with postal code	PK id UK city_id+name FK city_id	26	48	2	Reference	addresses
insurance_companies	Insurers (principals): contact, commission rate, remittance attributes	PK id UK code JSONB attrs	13	48	2	Reference	masters
policy_types	Policy types per product	PK id FK product_id JSONB attrs	13	32	1	Reference	masters
postal_codes	Postal code lookup (province, city, district)	PK id UK country_code+code+district	26	64	3	Reference	addresses
products	Insurance products / lines	PK id UK code JSONB attrs	8	48	2	Reference	masters
signatories	Authorised signatories for documents	PK id JSONB attrs	3	32	1	Reference	masters
states	Provinces / states	PK id FK country_id JSONB attrs	17	32	1	Reference	masters
vehicle_brands	Vehicle makes	PK id UK name JSONB attrs	8	48	2	Reference	quotations
vehicle_models	Vehicle models per make	PK id FK brand_id JSONB attrs	19	32	1	Reference	quotations
vehicle_variants	Vehicle variants (body type, seating)	PK id FK model_id JSONB attrs	20	32	1	Reference	quotations

Product configurator (4 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
product_components	Template components: covers, rating tables, rules, documents (JSONB data)	PK id FK template_id JSONB data	48	104	3	Reference	product-configurator
product_risk_mappings	Product / line-of-business risk definitions	PK id UK product_code JSONB configuration	8	48	2	Reference	product-configurator
product_risk_sections	Risk sections of a mapping with default rates	PK id UK mapping_id+section_code FK mapping_id	4	48	2	Reference	product-configurator
product_templates	Versioned product templates (config JSONB, rates, limits, insurers)	PK id FK parent_id, product_id JSONB config, features, insurers	13	80	2	Reference	product-configurator

Sales and policy (6 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
clients	Clients (insured parties) with KYC fields and owner	PK id UK client_code FK lead_id, owner_user_id JSONB extra	22	64	3	Transaction	clients
endorsements	Policy endorsements and cancellations, premium delta, billing link	PK id UK endorsement_number FK client_id, policy_id, receivable_id JSONB changes, endorsement_type_ids, completion	6	96	3	Transaction	endorsements
leads	Leads / prospects (individual or corporate), owner, status	PK id UK lead_number FK client_id, owner_user_id, product_id JSONB extra	18	80	4	Transaction	leads
policies	Issued policies: term, premium, insurer, billing mode, details / doc JSONB	PK id UK policy_number FK client_id, insurance_company_id, lead_id, owner_user_id +4 JSONB details, doc	31	176	6	Transaction	policies
policy_payments	Premium payments captured on a policy, awaiting finance confirmation	PK id FK confirmed_by, policy_id, receipt_id, receivable_id +2	1	64	3	Financial	policies
quotes	Quotations: vehicle / coverage JSONB, premium, taxes, commission, approval token hash	PK id UK quote_number FK agent_user_id, client_id, insurance_company_id, lead_id +4 JSONB vehicle, coverage, doc	25	208	5	Transaction	quotations

Claims (3 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
claim_field_changes	Field-level audit trail of claim edits	PK id FK claim_id	63	80	2	Log / audit	claims
claim_history	Claim status history	PK id FK claim_id	54	48	2	Log / audit	claims

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
claims	Claims: loss details, estimate, approved and settled amounts, handler, lifecycle status	PK id UK claim_number FK client_id, handler_user_id, policy_id JSONB details, driver, third_party	17	152	6	Transaction	claims

Renewals (7 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
renewal_activities	Contact and follow-up activities on a renewal	PK id FK renewal_id JSONB details	11	48	2	Log / audit	renewals
renewal_batch_policies	Policies in a renewal batch and their notice status	PK id UK batch_id+policy_id FK batch_id, policy_id, renewal_id	7	48	2	Transaction	renewals
renewal_batches	Batch renewal notice runs	PK id UK batch_number JSONB criteria	2	48	2	Transaction	renewals
renewal_notices	Renewal notices sent (stage, method, recipient, status)	PK id UK renewal_id+stage FK renewal_id	19	48	2	Log / audit	renewals
renewal_quotes	Renewal quotations (loading, discount, variance)	PK id UK quote_number FK renewal_id JSONB taxes, rating	2	64	3	Transaction	renewals
renewals	Renewal cases of expiring policies (status, notices, contact, premium old / new)	PK id FK client_id, new_policy_id, policy_id JSONB coverage_details, accessories, order_summary	15	112	4	Transaction	renewals
winback_campaigns	Win-back campaigns for lapsed business	PK id UK campaign_number JSONB offers	1	48	2	Transaction	renewals

Billing, receipts and collections (6 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
collection_actions	Collection actions: calls, e-mails, commitments	PK id FK collection_id	8	48	2	Log / audit	collections
collection_items	Collection follow-up items for open bills (commitment, escalation, assignee)	PK id UK receivable_id FK assigned_to, client_id, policy_id	32	48	2	Transaction	collections
receipt_applications	Application of receipt lines to bills, with journal link	PK id FK journal_id, receipt_id, receipt_line_id, receivable_id	28	48	2	Financial	receipts
receipt_lines	Receipt lines per policy (premium, taxes, EWT, discounts)	PK id UK receipt_id+line_no FK policy_id, receipt_id	28	48	2	Financial	receipts
receipts	Official receipts (OR-) header: payer, amount, mode, status	PK id UK receipt_number FK bank_id, client_id, policy_id, receivable_id	28	128	5	Financial	receipts
receivables	Premium bills (INV-) per policy / endorsement / renewal with balance and ageing	PK id UK bill_number FK booking_jv_id, client_id, policy_id	32	112	4	Financial	receipts

General ledger (5 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
accounting_periods	Accounting period open / closed status	PK period	0	16	1	Financial	accounting
entry_matches	Open-item matching of debit and credit journal lines	PK id FK credit_line_id, debit_line_id	0	32	3	Financial	accounting
gl_accounts	Chart of accounts (type, statement group, normal balance, open-item flag)	PK code FK parent_code	128	128	2	Reference	accounting
journal_lines	Journal lines: account, debit / credit, branch, department, client / policy tags	PK id FK client_id, jv_id, policy_id	286	152	4	Financial	journal-vouchers, accounting
journal_vouchers	Journal vouchers (JV-): source, status, maker-checker, reversal / correction links	PK id UK jv_number FK client_id, correction_of, policy_id, reversal_of +1	117	192	5	Financial	journal-vouchers, accounting

Disbursements and petty cash (9 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
checkbooks	Cheques issued (instrument, approval, printing)	PK id FK disbursement_id, invoice_list_id, journal_id	4	48	2	Financial	disbursements
disbursements	Payment vouchers (PV-) to insurers, agents, clients; approval and payment	PK id UK voucher_number FK bank_id, client_id, insurance_company_id, journal_id +1	23	112	4	Financial	disbursements
invoice_lists	Invoice lists (IL-) settled by a disbursement	PK id UK invoice_number FK client_id, disbursement_id, insurance_company_id, policy_id +1	6	64	3	Financial	disbursements
petty_cash_disbursements	Petty cash payments (VAT, WHT, journal)	PK id UK transaction_number FK fund_id, journal_id, request_id	1	48	2	Financial	payments
petty_cash_funds	Petty cash funds (size, limits, custodian, available cash)	PK id UK code FK custodian_user_id, journal_id	2	48	2	Financial	payments
petty_cash_receipts	Receipts into a petty cash fund	PK id UK receipt_number FK fund_id, journal_id	0	24	2	Financial	payments
petty_cash_replenishments	Fund replenishments from bank	PK id UK transaction_number FK fund_id, journal_id	0	24	2	Financial	payments
petty_cash_request_lines	Lines of a petty cash request	PK id FK request_id	3	32	1	Financial	payments
petty_cash_requests	Petty cash requests and approval	PK id UK request_number FK fund_id, requester_user_id	2	48	2	Financial	payments

Commission (2 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
commission_referrers	Agents / referrers: hierarchy, WHT rate, bank account	PK id FK parent_referrer_id, user_id	10	32	1	Reference	commission

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
commissions	Commission lines per policy / endorsement and referrer: accrual, eligibility, approval, payment	PK id FK accrual_jv_id, agent_user_id, disbursement_id, endorsement_id +4	53	120	3	Financial	commission, policies

Remittance and direct bill (9 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
commission_debit_note_collections	Insurer payments against a debit note (cash, EWT, Form 2307)	PK id UK collection_number FK debit_note_id, journal_id	2	64	3	Financial	remittance (direct bill)
commission_debit_note_lines	Policies / items on a debit note	PK id FK debit_note_id, item_id, policy_id	1	48	2	Financial	remittance (direct bill)
commission_debit_notes	Commission debit notes (DN-) to insurers for direct-billed policies	PK id UK dn_number FK insurance_company_id	1	64	3	Financial	remittance (direct bill)
direct_bill_items	Commission receivable items booked for direct-billed policies	PK id FK booking_jv_id, debit_note_id, endorsement_id, insurance_company_id +2	1	64	3	Financial	remittance (direct bill)
remittance_approvals	Multi-level approval requests with SLA and history	PK id FK action_by, delegated_to, initiator_id JSONB history	21	104	3	Log / audit	remittance
remittance_delegations	Approval delegations (period, transaction types, amount limit)	PK id FK delegate_id, delegator_id JSONB trans_types	1	32	1	Reference	remittance
remittance_items	Remittance work items (settlements, adjustments, transfers, exceptions)	PK id UK reference_no FK created_by, insurance_company_id, remittance_id JSONB data	25	96	3	Financial	remittance
remittance_lines	Policies included in a remittance	PK id FK policy_id, remittance_id	35	32	1	Financial	remittance
remittances	Premium remittances / bills to insurers (REM-): gross, commission, net due, approval, settlement	PK id UK remittance_number FK agent_user_id, insurance_company_id JSONB delivery_method, data	18	64	3	Financial	remittance

Reinsurance (7 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
cessions	Cessions of policies to treaties / facultative reinsurers	PK id FK facultative_reinsurer_id, policy_id, treaty_id	13	48	2	Transaction	reinsurance
reinsurance_bordereaux	Premium / claims bordereaux (entries JSONB, file)	PK id UK reference FK reinsurer_id, treaty_id JSONB totals	2	48	2	Transaction	reinsurance
reinsurance_exceptions	Reconciliation exceptions and resolution	PK id FK reconciliation_id	2	32	1	Log / audit	reinsurance
reinsurance_reconciliations	Reinsurer statement reconciliations	PK id UK reference FK reinsurer_id	2	48	2	Financial	reinsurance

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
reinsurance_recoveries	Claim recoveries from reinsurers	PK id UK recovery_number FK cession_id, claim_id, treaty_id JSONB cash_call, documents	6	48	2	Financial	reinsurance
reinsurance_treaties	Treaties (type, capacity, share, retention, approval)	PK id JSONB reinsurer_ids, terms	6	48	2	Transaction	reinsurance
reinsurers	Reinsurers (rating, capacity, contact)	PK id JSONB contact	6	48	2	Reference	reinsurance

Incentive (3 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
incentive_calculations	Incentive calculation batches with maker-checker and payment	PK batch_id FK approved_by, created_by, rejected_by, submitted_by JSONB programs_included	4	32	1	Financial	incentive
incentive_programs	Agent incentive programmes (metric, targets, tiers, eligibility)	PK id JSONB applicable_to, structure, eligibility	4	48	2	Reference	incentive
incentive_results	Per-agent incentive results and payouts	PK id UK program_id+agent_user_id+period FK agent_user_id, calculation_id, program_id	18	64	3	Financial	incentive

Reporting (3 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
generated_reports	Generated report files (code, parameters, format, storage key, rows)	PK id JSONB params, totals	115	144	3	Temporary	reports
report_definitions	Report catalogue (parameters, query, columns, roles, permission)	PK code JSONB parameters, default_columns	19	112	1	Reference	reports
report_schedules	E-mailed report schedules (cron, format, recipients)	PK id FK job_id, report_code JSONB params	0	16	1	Reference	reports

4 Other database objects

4.1 Database facts

Item	Value (local test database)
PostgreSQL version	16.13 (Ubuntu 16.13-0ubuntu0.24.04.1)
Database size	20.2 MB (test data)
Server time zone	Etc/UTC
Tables (schema public)	102
Columns by type	text 952, timestamp with time zone 233, numeric 178, integer 89, jsonb 77, date 64, bigint 30, boolean 21, ARRAY 2
Indexes	226 (161 unique incl. 102 primary keys, 15 partial, 1 GIN)
Foreign keys	174 (128 without a supporting index)
Identity sequences (serial / identity columns)	46
Document-number counters (table sequences)	32 counters for 2026
Extensions	pgcrypto 1.3, plpgsql 1.0
Application functions	gl_accounts_defaults, jv_check_posting, next_number, numbering_prefix, rpt_age_bucket, set_updated_at
Triggers	gl_accounts_defaults_trg (gl_accounts), jv_check_posting_trg (journal_vouchers), roles_updated (roles), users_updated (users)
Views / materialised views	none
Migrations applied	32 (0001_core.sql to 0110_brokerverse_ootb_branding.sql)
Application settings (app_settings)	273 keys in 31 groups
Scheduled jobs (scheduled_jobs)	9
Snapshot taken	2026-09-29T09:25:28+00:00

4.2 Indexes

226 indexes: 102 primary keys, 59 further unique indexes, 15 partial indexes and 1 GIN index (`master_records_data_idx` on `master_records.data`). Partial indexes: `policies_billing_mode_idx` (direct-billed policies), `direct_bill_items_open_idx` (unbilled items), `refresh_tokens_user_idx` (live tokens), `clients_lead_idx`, `renewals_open_policy_uq`, `entry_matches_debit_idx`, `entry_matches_credit_idx`, `commissions_policy_referrer_uq`, `master_records_code_uq`, `product_templates_code_uq`, `product_components_code_uq`, `remittance_approvals_open_uq`, `reinsurance_treaties_number_uq`, `cessions_number_uq`, `incentive_programs_code_uq`.

Foreign keys without a supporting index

Gap: 128 of the 174 foreign keys have no index whose first column is the foreign-key column. The table below lists them; the ones on growing transactional and log tables should be indexed (document 02, section on constraints and indexes, and document 06).

Table	Foreign-key columns without an index whose first column is the FK column
agent_events	user_id

Table	Foreign-key columns without an index whose first column is the FK column
cessions	facultative_reinsurer_id, policy_id, treaty_id
checkbooks	disbursement_id, journal_id
cities	state_id
claims	handler_user_id
clients	owner_user_id
collection_items	assigned_to, client_id, policy_id
commission_debit_note_collections	journal_id
commission_debit_note_lines	item_id, policy_id
commission_referrers	parent_referrer_id, user_id
commissions	accrual_jv_id, agent_user_id, disbursement_id, endorsement_id, payment_jv_id, quote_id
coverages	policy_type_id
direct_bill_items	booking_jv_id, debit_note_id, endorsement_id, reversal_jv_id
disbursements	bank_id, client_id, insurance_company_id, journal_id, policy_id
endorsements	client_id, receivable_id
gl_accounts	parent_code
incentive_calculations	approved_by, created_by, rejected_by, submitted_by
incentive_results	calculation_id
invoice_lists	client_id, disbursement_id, insurance_company_id, policy_id, receipt_id
job_runs	job_id
journal_lines	policy_id
journal_vouchers	client_id, correction_of, reversal_of, reversed_by_jv
leads	client_id, owner_user_id, product_id
password_resets	user_id
petty_cash_disbursements	fund_id, journal_id, request_id
petty_cash_funds	custodian_user_id, journal_id
petty_cash_receipts	fund_id, journal_id
petty_cash_replenishments	fund_id, journal_id
petty_cash_request_lines	request_id
petty_cash_requests	fund_id, requester_user_id
policies	insurance_company_id, lead_id, owner_user_id, policy_type_id, product_id, renewed_from
policy_payments	confirmed_by, receipt_id, receivable_id, rejected_by, submitted_by
policy_types	product_id
product_components	template_id
product_templates	parent_id, product_id
quotes	agent_user_id, client_id, insurance_company_id, policy_id, policy_type_id, product_id, signatory_id
receipt_applications	journal_id, receipt_id, receipt_line_id
receipt_lines	policy_id
receipts	bank_id, client_id, receivable_id
receivables	booking_jv_id
reinsurance_bordereaux	reinsurer_id, treaty_id
reinsurance_exceptions	reconciliation_id
reinsurance_reconciliations	reinsurer_id
reinsurance_recoveries	cession_id, claim_id, treaty_id
remittance_approvals	action_by, delegated_to, initiator_id
remittance_delegations	delegate_id, delegator_id

Table	Foreign-key columns without an index whose first column is the FK column
remittance_items	created_by, insurance_company_id, remittance_id
remittance_lines	policy_id, remittance_id
remittances	agent_user_id, insurance_company_id
renewal_batch_policies	policy_id, renewal_id
renewals	client_id, new_policy_id
report_schedules	job_id, report_code
role_permissions	permission_id
states	country_id
user_roles	role_id
vehicle_models	brand_id
vehicle_variants	model_id

Log tables and their indexes

Table	Existing indexes	Observation
audit_log	PK; (entity, entity_id)	The audit trail endpoint (GET /api/settings/audit) filters by entity, entity id or username and returns at most 1,000 rows, newest first; it has no date filter. A filter by username and any time-based review or purge scan the whole table. Recommended: index (at) and (username, at), and a date-range filter for audit reviews.
login_history	PK; (user_id, at DESC); (at DESC)	Adequate.
notifications	PK; (user_id, is_read, created_at DESC)	Adequate for the per-user bell; notifications are never purged (document 10).
email_outbox	PK only	The outbox job selects <code>status = 'queued' AND attempts < 5 ORDER BY id LIMIT 50</code> every 5 minutes; without an index on <code>status</code> this scans the whole table as it grows. Recommended: partial index <code>WHERE status = 'queued'</code> .
job_runs	PK only	About 1,735 rows a day (renewal-queue every minute, e-mail outbox every 5 minutes, 7 daily jobs); queried by <code>job_id</code> and <code>started_at</code> on every scheduled run and by the history screen. Recommended: index (<code>job_id, started_at DESC</code>) and a retention job.
refresh_tokens	PK; live tokens per user (partial); family	Expired and revoked rows are never deleted.
generated_reports	PK; (created_at); (code)	Purged after <code>reports.retention_days</code> by the daily-reports job.

4.3 Sequences

46 identity sequences (<table>_id_seq) back the serial primary keys of the reference and log tables. Document numbers do not use PostgreSQL sequences but the counter table [sequences](#) (document 02).

4.4 Extensions, functions and triggers

Object	Detail
Extensions	plpgsql 1.0 (built in), pgcrypto 1.3 (gen_random_bytes() for primary-key defaults)
next_number(name, prefix, width)	Allocates the next document number of the year from sequences
numbering_prefix(entity, default)	Reads numbering.<entity>.prefix from app_settings
rpt_age_bucket(age, buckets)	Ageing bucket label for reports
jv_check_posting() / trigger jv_check_posting_trg on journal_vouchers	Refuses posting an unbalanced journal or one in a closed period
gl_accounts_defaults() / trigger gl_accounts_defaults_trg on gl_accounts	Derives the normal balance of an account from its type
set_updated_at() / triggers users_updated, roles_updated	Maintain updated_at on users and roles

4.5 Application settings by group

app_settings holds 273 configuration keys in 31 groups (reference data, seeded from backend/src/db/seeds/settings.json and SQL seed files, editable in Master > Configuration and System Settings).

Group	Keys	Group	Keys	Group	Keys
accounting	21	general	11	quote	4
branding	6	incentive	5	reinsurance	6
claims	16	leads	5	remittance	12
collections	7	limits	7	renewals	23
commission	8	notification	5	reports	17
currency	4	numbering	43	security	18
dashboard	3	policies	3	system	1
direct_bill	10	policy	4	tax	5
email	5	premium	3	uploads	4
endorsements	1	product	4		
finance	5	quotations	7		
Total	273				

Settings relevant to operations: **security.*** (password policy, rate limits, 2FA roles, scoped roles), **limits.*** (lockout, idle time-out, quote validity, ageing buckets, renewal notice days), **notification.*** (e-mail on / off, sender), **reports.retention_days** (90), **reports.download_link_ttl_hours** (72), **reports.max_rows** (50,000), **uploads.*** (allowed types and sizes), **general.timezone** (Asia/Manila), **numbering.*** (43 document prefixes).

4.6 Scheduled jobs

9 rows in `scheduled_jobs` (document 01, section on scheduled jobs). Report schedules add rows with the code `report -<id>`.

5 Related documents

This document is one of eleven that together form the BrokerVerse Solution Architecture. They are maintained and issued together (source and build: [docs/architecture/README.md](#)).

No.	Document	File
01	Solution Component Diagram and Application Architecture	01_BrokerVerse_Solution_Component_Diagram_Application_Architecture.pdf
02	Database Design	02_BrokerVerse_Database_Design.pdf
03	Database Inventory (this document)	03_BrokerVerse_Database_Inventory.pdf
04	Technology Stack	04_BrokerVerse_Technology_Stack.pdf
05	Shared Service Components	05_BrokerVerse_Shared_Service_Components.pdf
06	Capacity and Performance	06_BrokerVerse_Capacity_and_Performance.pdf
07	High Availability and Resiliency	07_BrokerVerse_High_Availability_and_Resiliency.pdf
08	RTO and RPO	08_BrokerVerse_RTO_and_RPO.pdf
09	Backup and Recovery	09_BrokerVerse_Backup_and_Recovery.pdf
10	Data Archival, Housekeeping and Restoration	10_BrokerVerse_Data_Archival_Housekeeping_and_Restoration.pdf
11	Monitoring	11_BrokerVerse_Monitoring.pdf

6 Glossary

Terms and abbreviations used in this document.

Term	Meaning
Audit log / audit trail	Table audit_log: who changed which record, when, from which IP, with the before and after values.
bcrypt	Adaptive password-hashing function (bcryptjs library) used for user passwords.
Cession	The share of a policy's risk and premium passed to a reinsurer under a treaty or facultatively.
Cron	Time-based schedule expression (minute hour day month weekday) used by the scheduled jobs.
Debit note (DN)	Commission debit note raised by the broker to an insurer for direct-billed policies.
Direct bill	Billing mode in which the insurer collects the premium and the broker bills the insurer for its commission.
Endorsement	Change to an issued policy (details, cover, premium or cancellation).
EWT	Expanded Withholding Tax (Philippines), e.g. withheld by an insurer on commission (BIR Form 2307).
FK	Foreign key: column that references the primary key of another table.
GIN	Generalised inverted index (PostgreSQL), used for JSONB searches.
HMAC	Keyed-hash message authentication code; signs file links and hashes one-time codes.
JSONB	PostgreSQL binary JSON column type, used for flexible document parts of records.
JV	Journal voucher: balanced set of debit and credit ledger lines.
KYC	Know your customer: identification data required before a policy is issued.
Maker-checker	Control in which a record created or submitted by one user must be approved by a different user.
Migration	Numbered SQL file in backend/src/db/migrations applied once, in order, on API start.
OR	Official receipt issued for premium received.
PK	Primary key: column(s) that uniquely identify a row.
PV	Payment voucher (disbursement) to an insurer, agent, client or supplier.
Receivable / bill	Premium bill (INV-) raised when a policy, endorsement or renewal is issued.
Remittance	Payment of collected premium, net of commission, from the broker to the insurer.
RPO	Recovery point objective: the maximum acceptable data loss, measured in time.
RTO	Recovery time objective: the maximum acceptable time to restore service.
Seed	Idempotent data load run on every API start: reference data always, sample data only when SEED_SAMPLE_DATA is on.
SLA	Service level agreement / target time, e.g. for remittance approvals.
Token version	users.token_version, carried in every access token; raising it ends all sessions of the user.
TOTP / 2FA	Time-based one-time password (RFC 6238) used for two-factor authentication.
Treaty	Reinsurance contract under which risks are ceded automatically within agreed limits.
UAT	User acceptance testing environment.
VAT	Value-added tax (12 % in the Philippines).
WHT	Withholding tax deducted from payments such as agent commission.